Syamprasad

Software Developer

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PROFESSIONAL SUMMARY

Computer Vision Engineer with 5+ years of experience building and deploying deep learning pipelines for object detection, tracking, segmentation, OCR, pose estimation, and real-time video analytics. Skilled in optimizing YOLO models with ONNX and TensorRT for edge deployment (NVIDIA Jetson) and integrating them with backend services, REST APIs, and scalable cloud environments (AWS).

WORK EXPERIENCE

Freelance Developer & Technical Mentor

Independent

May 2025 - Present

- Deliver Python, machine learning, and computer vision projects from requirements to working solutions.
- Mentor learners and run workshops on Python, machine learning, and project development.

Python Developer (Computer Vision & ML)

Gama Security Solution, Dubai

Sep 2022 - May 2025

- Built YOLO-based analytics on CCTV/RTSP streams, achieving over 94% detection accuracy across complex lighting conditions for people/vehicle detection, entry/exit counting with ID tracking, occupancy alerts, fence-breach detection, and PPE compliance.
- Optimized models for latency, memory, and real-time inference, boosting detection accuracy to 92%+ mAP; deployed TensorRT models on NVIDIA Jetson AGX for low-power edge operation with live heatmaps.
- Developed face recognition and FAISS-based face search for biometric security and access control.
- Implemented YOLOv8 pose-based fall detection and ANPR with OCR on live camera feeds.

Python Developer (Computer Vision, ML & Web)

Nacre Systems, Palakkad

May 2021 - May 2022

- Developed computer vision and ML solutions for video analytics (detection, segmentation, real-time pipelines) and Python web applications.

Python Developer (Computer Vision, ML & Web)

Grapesgenix Technical Solutions, Thrissur

Nov 2020 - May 2021

- Developed video analytics, ML, and web applications using Python, Django, and Flask.

CORE SKILLS

Computer Vision

OpenCV, Object Detection, Tracking, Segmentation, OCR, Pose Estimation, Face Recognition, FAISS, Video Processing

Deep Learning

YOLO (YOLOv8), PyTorch, TensorFlow, Keras, CNN, Transfer Learning, Custom Model Training

Cloud & Edge AI

AWS, ONNX, TensorRT, NVIDIA Jetson AGX / Nano, Real-Time Inference, RTSP/CCTV Streams

Programming

Python, C++, NumPy, Pandas, Scikit-learn, Django, Flask, REST APIs, Git, MySQL, ClickHouse

KEY PROJECTS

Smart Surveillance (Jetson AGX): YOLO detection, tracking, heatmaps, and TensorRT edge deployment.

ANPR with OCR: License plate recognition from real-time camera streams.

FAISS Face Search: Embedding-based retrieval, robust to masks and lighting changes.

Nail Segmentation: YOLO segmentation for virtual nail try-on.

AI Job Agent: AI-powered resume-to-job matching system.

EDUCATION

Bachelor of Computer Applications (BCA)

2018 - 2021

Python Django Full Stack Course

2020

LANGUAGES

Malayalam, English, Tamil, Hindi